Physics-informed identification of coupled beam–RF feedback dynamics from closed-loop measurements in a storage ring

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The fundamental-mode RF system of an electron storage ring and the stored beam form one coupled dynamical system whose behavior changes with beam current and with the settings of the RF feedback. We ask whether closed-loop measurements made during operation can identify a compact model that predicts these changes. From eighteen closed-loop transfer-function records of the SPEAR3 RF station, taken with its existing noise-injection routine at beam currents from 20 to 500 mA and at several loop gains and phases, we identify a physics-informed two-quadrature model of beam, cavities, klystron and feedback. Single records, measured with a one-dimensional excitation, do not determine the klystron's response asymmetry. Records at several loop settings, together with the band of the coupled beam mode, determine its compression but leave its AM/PM conversion correlated with the loop phase; records frozen at trips of the same klystron supply it ($k_{\mathrm{AP}} = -0.10 \pm 0.05$). Of the competing explanations for the large change of the response with beam current—effective linear parameters, unequal cavity detunings, beam dynamics at fixed hardware, and a compressed klystron—only the last describes all records with one set of station constants ($\chi^2/\nu = 1.016$ with 45 parameters for 25200 data points), and these constants agree with independent station data. The klystron's compression, its amplitude gain relative to its phase gain, falls from 0.56 at 22 mA to 0.31 at 500 mA as the tube, held at a fixed drive, works closer to saturation; at full output the trip records give 0.34–0.42. The model predicts records at beam currents it was not trained on almost at in-sample quality, where interpolated effective parameters fail. Its out-of-band prediction of the coupled mode isolates the synchrotron focusing as the one beam parameter the loop band cannot determine; identified from the beam band, it implies an incoherent synchrotron frequency 9% above the design value, a prediction that a low-current measurement can test.

I. INTRODUCTION

In an electron storage ring the fundamental-mode RF system and the stored beam form one coupled dynamical system. The beam induces voltage in the accelerating cavities; the cavities are detuned to compensate the reactive part of that loading; feedback around the klystron lowers the impedance that the beam sees; and the coherent dipole motion of the bunches responds to every perturbation of the gap voltage that the feedback leaves. The classical analyses—Robinson's stability criteria [1], Pedersen's beam-loaded cavity [2], and the treatments of RF feedback by Boussard [3] and Wilson [4]—describe this coupling with a handful of parameters: cavity bandwidth and detuning, loop gain, phase and delay, beam current and synchrotron frequency. Detailed models built on the same physics were used to design and commission the low-level RF (LLRF) systems of PEP-II and the LHC [5, 6].

At an operating machine, however, several of these parameters are not known well. Components age and are replaced, calibrations are uncertain, parts of the loop setting are not recorded, and the klystron runs close to saturation, where its small-signal response to a change of drive amplitude differs from its response to a change of drive phase. A model assembled from design values may describe one operating point and fail at another. The operational questions—how the damping and frequency of the coherent mode, and the transfer of RF noise to the beam, change with beam current, loop gain and loop phase—call for a model identified from measurements. Such measurements must be made on a running machine: with the feedback closed, with stored beam, with small excitation, and preferably with the diagnostics that already exist [7, 8].

This paper asks whether operational closed-loop measurements can identify a compact model that predicts beam–cavity–amplifier–feedback interactions across operating conditions. The approach is physics-informed (grey-box) identification [9, 10]. Physics supplies the structure—cavity resonators, rigid-bunch dipole dynamics, the loop topology and the controller networks, the steady-state beam loading, the way the loop settings enter—and the data determine, and test, the parameters that the physics leaves uncertain. The emphasis is not on fitting one transfer function, which many models do, but on whether one set of physical parameters accounts for the changes of the dynamics as the beam current, the loop gain and the loop phase change, and whether the identified model predicts quantities that were not used to identify it.

The data are the closed-loop responses recorded by the existing routine of the SPEAR3 RF station—band-limited noise injected into the klystron drive and the feedback signal recorded—at beam currents from 20 to 500 mA and at several loop gains and phases, on two days a month apart, together with records that the station froze at trips of its present klystron. Four features make the problem a stringent test: (i) the excitation is one-

dimensional, the same noise sequence on both quadratures, so each record measures a single projection of a response that is not rotation-invariant once the klystron is compressed; (ii) part of the loop phase is not recorded, because a tracking correction drifts by several degrees; (iii) the beam's coherent mode occupies a few kilohertz around $\pm f_s$, a band the routine itself discards; and (iv) independent information on the klystron exists only in those trip records, which are five years older than the closed-loop data.

The paper makes four contributions. Section III gives a two-quadrature (widely linear [11]) model of the closed loop with beam, including the klystron's compression and AM/PM conversion, and derives what a one-dimensional excitation records. Section IV shows that single records do not determine the klystron's asymmetry, that records taken at different loop settings determine its compression but not its AM/PM conversion, and that the trip records supply the latter. Section V tests competing explanations for the change of the response with beam current. Section VI validates the selected model by out-of-band prediction of the coupled beam mode, by prediction across operating conditions, and against the klystron compression measured in the trip records; the out-of-band test exposes, and the beam band then identifies, the one beam parameter that the loop band cannot determine—the synchrotron frequency.

II. THE MEASURED SYSTEM

A. RF station and direct feedback

SPEAR3 is a 3 GeV storage ring with harmonic number $h = 372$ and $f_{\mathrm{RF}} = 476.3\,\mathrm{MHz}$, operated at 500 mA in top-off [12, 13]. One 1.5 MW klystron drives four single-cell PEP-II cavities [14] through a circulator and a waveguide splitter; together they provide $V_{\mathrm{RF}} = 2.85\,\mathrm{MV}$. Table I lists the parameters used. The cavities are overcoupled ($\beta \approx 3.8$, loaded quality factor $Q_L \approx 6800$, half-bandwidth 35 kHz) and are detuned below f_{RF} in proportion to the beam current, by about 36 kHz at 500 mA, to cancel the reactive beam loading; the tuner loops hold the load angle, so the detuning follows the current.

The legacy LLRF processor (RFP), an analog VXI module of PEP-II design [15], regulates the vector sum of the four probe signals with a direct loop: the measured vector sum is rotated and scaled by a baseband modulator (settings `PHASE.C` and `COUNTS.A`, with a frequency-offset tracking correction added to the phase), the reference is subtracted, and a lead network and a lag (integral) network shape the error before the I/Q modulator that drives the klystron (Fig. 1).

The loop delay is about 0.63 μs, mostly in the klystron and the cables to the tunnel. A digital ripple loop acting on the 60-Hz harmonics up to 2.9 kHz, and slow EPICS loops for the klystron voltage, the cavity tuners and the gain tracking, complete the system. The klystron-voltage

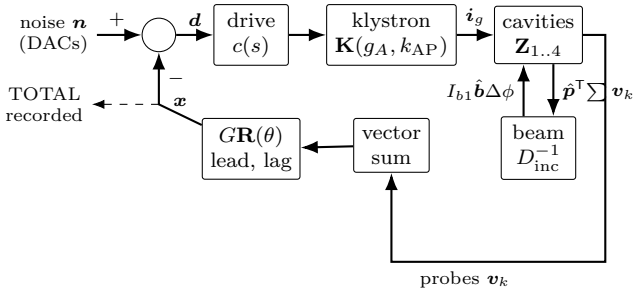


FIG. 1. The measured system. Band-limited noise $\boldsymbol{n}$ is added to the klystron drive after the controller; the controller output $\boldsymbol{x}$ (TOTAL) is recorded. The beam couples to the four cavities through its dipole motion $\Delta\phi$ [Eq. (2)]; the klystron responds differently to drive amplitude and phase [Eq. (5)]. Bold symbols are two-quadrature vectors and matrices in the cavity frame.

TABLE I. Parameters of the SPEAR3 RF station used in the model. Fixed values are taken from design documents and station measurements [13, 16]; identified values are those of Sec. V.

Quantity	Symbol	Value
Energy; harmonic number	E_0; h	3.0 GeV; 372
RF frequency	f_{RF}	476.304 MHz
Energy loss per turn (measured)	U_0	1.09 MeV
Total gap voltage	V_{RF}	2.85 MV
Synchronous phase (from crest)	ϕ_s	67.5°
Radiation damping rate	α_{rad}	$424\,\mathrm{s}^{-1}$
Momentum compaction (design)	α_c	1.18×10^{-3}
Cavities; R/Q each (circuit)	n_{cav}, R/Q	4; 116.6
Lead network zero; pole		944 kHz; 2.39 MHz
Lag network zero; pole		3.19 kHz; 0.26 kHz
Drive-chain pole	$\omega_a/2\pi$	2.5 MHz
Klystron drive (set; saturation)	P_{d}	28 W; 80 W
Klystron AM/PM (trip records)	k_{AP}	−0.10

loop holds the drive power at a set point, 28 W, 4.4 dB below the saturation setting of 80 W, by moving the HVPS voltage, so the klystron's voltage, and with it its gain, change with the beam current. In the snapshots of 2026 the drive is 28 W without beam and 29 W at 500 mA, with the HVPS at 67 kV and 75 kV, a klystron gain of 41.5 dB and 44.4 dB, and an output of 398 kW and 790 kW at an efficiency of 32 % and 47 %: at the same drive the tube works much closer to saturation with beam than without. Gain tracking holds the product of a DSP coefficient and the measured klystron gain constant; in the same snapshots it is 107.3 at 500 mA and 107.0 without beam, although the klystron gain differs by 2.9 dB. The forward-path gain is therefore independent of the operating point, and the loop gain changes only with `COUNTS.A`. Neither the closed-loop records nor the trip records carry the HVPS voltage or the klystron's drive and output power.

B. The closed-loop measurement

The RFgui routine "Measure Direct Closed" plays a stored sequence of band-limited noise through two DACs whose output is summed into the klystron drive after the controller, and records the controller output (the summed feedback signal, TOTAL) in the RFP signal memory: 2^{18} complex samples at eight samples per turn (10.24 MS/s, 25.6 ms). The stored reference is flat to ±500 kHz. The routine estimates the transfer function from the reference to the record, which for a complex-linear loop is the complementary sensitivity $-L/(1+L)$, and discards ±18.8 kHz around the carrier. The same sequence is stored for the I and Q DACs (correlation 0.9998), so the excitation is one-dimensional. The measurement is non-invasive: the injected noise is a fraction of a percent of the drive and the routine runs with stored beam.

C. Data

Table II lists the eighteen records of 2026 used in this paper. On 2 October the current was varied from 20 to 497 mA at fixed gain, and the phase setting was stepped at 22 mA. On 3 November, at about 500 mA, the phase setting was stepped from −30° to −35° and the gain setting from +1.3 to −0.2 dB, after one record at 242 mA. Each record file name carries the station voltage, the beam current, `COUNTS.A`, `PHASE.C` and the time, e.g. `v2850i494gm002pm0350d20261103172932` for 2.85 MV, 494 mA, −0.2 dB and −35.0°; the frequency-offset tracking correction was on and is not recorded (station snapshots show it drifting between 2° and 13°).

The trip records are a second, independent data set, of the same klystron, which has been in service since September 2020. At each trip the station freezes, together, the last 5.0 ms of the klystron drive I/Q (RFP signal memory), one cavity probe I/Q (RFP cavity memory), and the klystron output amplitude (IQ&A history), all at 10.24 MS/s. Of the 15 records of trips in 2021, 10 hold RF in every memory: 4 at full klystron output and 6 at 49% of that power—the output the station needs without beam (50% in the snapshots)—with the same drive to within 3%. They carry neither the beam current, nor the loop settings, nor the HVPS voltage.

D. Frequency-response estimation

For each record the transfer function $\hat{H}(f)$ is the Welch H_1 estimate [17] with Hann windows and 50 % overlap, in the convention of the routine (complex conjugate of the cross-spectral ratio), which makes the loop delay positive and places the high-current detuning below f_{RF}. Two resolutions are used: 8192-point segments (1.25 kHz) for the *wideband* $15 \le |f| \le 400$ kHz, and 2^{15}-point segments (312 Hz) for the *beam band* $3 \le |f| \le 16$ kHz.

TABLE II. Closed-loop records of 2026 (station voltage 2.85 MV). I_b: beam current; `COUNTS.A`: loop gain setting; `PHASE.C`: loop phase setting. Group: operating point used for the klystron and noise-angle parameters.

Date, time	I_b (mA)	`COUNTS.A` (dB)	`PHASE.C` (deg)	Group
2 Oct, 09:33	22	+0.9	−14	1
2 Oct, 09:41	22	+0.9	−17	1
2 Oct, 09:47	22	+0.9	−20	1
2 Oct, 09:49	22	+0.9	−23	1
2 Oct, 10:09	22	+0.9	−23	1
2 Oct, 13:44	363	+0.9	−23	2
2 Oct, 13:49	494	+0.9	−23	4
2 Oct, 13:51	497	+0.9	−20	4
2 Oct, 13:58	20	+0.9	−20	1
2 Oct, 17:20	395	+0.9	−20	3
3 Nov, 17:07	242	+1.3	−20	5
3 Nov, 17:22	497	+1.3	−30	6
3 Nov, 17:26	499	+1.3	−35	6
3 Nov, 17:27	497	+1.3	−35	6
3 Nov, 17:28	495	+0.3	−35	6
3 Nov, 17:29	494	−0.2	−35	6
3 Nov, 17:30	500	+0.0	−35	6
3 Nov, 17:31	498	+0.3	−35	6

The variance of each bin follows from the coherence γ^2, $\sigma_H^2 = |H|^2(1-\gamma^2)/(\gamma^2 n_{\mathrm{eff}})$ with n_{eff} the overlap-corrected number of segments; the output-noise and input spectra entering it are median-smoothed over neighbouring bins so that a bin's weight does not follow the noise of its own estimate. In the beam band, where the coupled mode is only a few bins wide, the model is convolved with the window's power kernel before it is compared with the estimate. The complex residuals $(H_{\mathrm{model}} - \hat{H})/\sigma_H$ define a Gaussian likelihood, so that $\chi^2/\nu \approx 1$ indicates a fit at the noise level and differences of χ^2 are log-likelihood ratios. A split-record test confirms the calibration of σ_H for the bulk of the bins (App. B); bins carrying interference lines are down-weighted by the smoothing, which is why the best fits reach χ^2/ν slightly below one (0.59–0.82 for record-by-record linear fits at 22 mA).

III. PHYSICS-INFORMED MODEL

A. Two-quadrature representation

Signals are complex envelopes on the carrier $e^{j\omega_{\mathrm{RF}}t}$, and s is the Laplace variable of the envelope ($s = j2\pi f$ at an offset f from f_{RF}). Because the klystron and the beam treat amplitude and phase differently, each envelope is written as the real vector $\boldsymbol{x} = (x_I, x_Q)^{\mathsf{T}}$ in the *cavity frame*, in which the steady total gap voltage is real, and each complex-coefficient transfer function $F(s)$

as the real matrix

$$\mathbf{F}(s) = \begin{pmatrix} F_r & -F_i \\ F_i & F_r \end{pmatrix}, \quad F_{r,i} = \tfrac{1}{2}\left[F(s) \pm \bar{F}(s)\right] / \{1, j\}, \tag{1}$$

with $\bar{F}(s) = \overline{F(\bar{s})}$ (Doc P, App. C [16]). Blocks with a real impulse response—delays, the lead and lag networks, the drive-chain pole—are scalars times the identity; a detuned cavity, a rotation, the compressed klystron and the beam are not.

B. Cavities and beam

Cavity k is the baseband resonator $Z_k(s) = R_L\omega_{1/2}/(s+\omega_{1/2}-j\Delta\omega_k)$, with $R_L = (R/Q)Q_L$, $\omega_{1/2} = \omega_{\rm RF}/2Q_L$ and $\Delta\omega_k$ its detuning from $f_{\rm RF}$. All four receive the same generator current $\boldsymbol{i}_g$ and the same beam current, so $\boldsymbol{v}_k = \mathbf{Z}_k(\boldsymbol{i}_g - I_{b1}\hat{\boldsymbol{b}}\,\Delta\phi)$, where $I_{b1} = 2I_b$ and $\Delta\phi$ is the phase of the bunch centroids (positive when late). For rigid bunches in the uniform-fill dipole mode [2, 18],

$$\left(s^2 + 2\alpha_{\rm rad}s + \omega_s^2\right)\Delta\phi = \kappa_1\,\hat{\boldsymbol{p}}^{\mathsf{T}}\textstyle\sum_k \boldsymbol{v}_k, \tag{2}$$

with $\kappa_1 = \alpha_c\omega_{\rm RF}/[(E_0/e)T_{\rm rev}]$, $\omega_s^2 = \kappa_1 V_{\rm RF}\sin\phi_s$, and the unit vectors $\hat{\boldsymbol{b}} = (-\sin\phi_s, -\cos\phi_s)^{\mathsf{T}}$, along which a phase shift changes the beam current, and $\hat{\boldsymbol{p}} = (\cos\phi_s, -\sin\phi_s)^{\mathsf{T}}$, along which the bunch senses the voltage. Eliminating $\Delta\phi$ gives the beam-loaded plant from generator current to the measured vector sum $\boldsymbol{v}_m = \sum_k w_k\boldsymbol{v}_k$ ($w_k = 1/4$):

$$\mathbf{P}_b = \mathbf{Z}_m - \frac{\kappa_1 I_{b1}}{D(s)}\,\mathbf{Z}_m\hat{\boldsymbol{b}}\,\hat{\boldsymbol{p}}^{\mathsf{T}}\mathbf{Z}_\Sigma, \tag{3}$$

$$D(s) = s^2 + 2\alpha_{\rm rad}s + \omega_s^2 + \kappa_1 I_{b1}\,\hat{\boldsymbol{p}}^{\mathsf{T}}\mathbf{Z}_\Sigma\hat{\boldsymbol{b}}, \tag{4}$$

where $\mathbf{Z}_\Sigma = \sum_k \mathbf{Z}_k$ and $\mathbf{Z}_m = \sum_k w_k\mathbf{Z}_k$. $D(s) = 0$ is Pedersen's quartic for the beam-loaded cavity without feedback [2].

C. Klystron

About its operating point the klystron is memoryless at these frequencies (its bandwidth is 5 MHz). A change of drive amplitude changes the output amplitude with a relative gain g_A and the output phase by $k_{\rm AP}$ radians per unit relative amplitude (AM/PM conversion [19, 20]); a change of drive phase passes with unit relative gain. In the cavity frame

$$\mathbf{K} = \mathbf{R}(\theta_g)\begin{pmatrix} g_A & 0 \\ k_{\rm AP} & 1 \end{pmatrix}\mathbf{R}(-\theta_g), \tag{5}$$

where $\mathbf{R}$ is a rotation and θ_g the angle of the steady generator current, computed from the beam loading [$\boldsymbol{i}_{g0} = V_{\rm cav}/Z_m(0) + I_{b1}e^{-j\phi_s}$]. A linear klystron has $g_A = 1$, $k_{\rm AP} = 0$; a klystron driven close to saturation has $g_A < 1$. Its phase-path gain is absorbed in the loop gain. Because the station holds the drive and moves the cathode voltage (Sec. II), the klystron's gain and its place on its transfer curve change with the beam current. Gain tracking removes the change of gain from the loop; the change of place, and with it of g_A and $k_{\rm AP}$, remains.

D. Loop and recorded response

The noise $\boldsymbol{n}$ enters the drive after the controller, and the record is the controller output $\boldsymbol{x}$:

$$\boldsymbol{x} = -\mathbf{M}(\boldsymbol{x} + \boldsymbol{n}), \qquad \mathbf{M}(s) = G\,c(s)\,\mathbf{R}(\theta)\,\mathbf{P}_b(s)\,\mathbf{K}/R_L, \tag{6}$$

where G is the loop gain of the phase path at an on-tune resonance, θ the loop phase error (positive = lead), and

$$c(s) = e^{-s\tau}\,\frac{1+s/\omega_z}{1+s/\omega_p}\,\frac{s+\omega_{iz}}{s+\omega_{ip}}\,\frac{\omega_a}{s+\omega_a} \tag{7}$$

collects the delay τ, the lead and lag networks and the drive-chain pole (Table I). The closed-loop response is $\mathbf{T} = -(\mathbf{1}+\mathbf{M})^{-1}\mathbf{M}$. A one-dimensional excitation along the angle α in the cavity frame records

$$H(f) = e^{-j\alpha}\begin{pmatrix}1 & j\end{pmatrix}\mathbf{T}\begin{pmatrix}\cos\alpha \\ \sin\alpha\end{pmatrix} = H_1(f) + H_2(f)\,e^{-2j\alpha}, \tag{8}$$

with the direct and image responses $H_1 = [T_{11} + T_{22} + j(T_{21} - T_{12})]/2$ and $H_2 = [T_{11} - T_{22} + j(T_{21} + T_{12})]/2$ [11, 21]. For a linear klystron and no beam, $H_2 = 0$ and $H = -L/(1+L)$ with $L = Ge^{j\theta}c\,Z_m/R_L$. Otherwise a single record measures one mixture of H_1 and H_2, set by α, which depends on the klystron's insertion phase but not on the loop settings, because the noise enters after the controller's rotation. Elements outside the loop (the DAC filters and the recording channel) multiply H by a smooth factor common to all records, $\exp(c_1x + c_2x^2)$ with $x = f/400\,\text{kHz}$; its linear phase is not separable from the loop delay and is set to zero.

E. The coupled mode

The poles of $\mathbf{T}$ are the zeros of $\det[\mathbf{1} + \mathbf{M}(s)]\,D(s)$. Tracked from zero current, one pair is the coherent dipole mode of beam, cavities, klystron and loop, $s = -\sigma \pm j\omega_c$. Without feedback it is Pedersen's root; with a linear klystron it is the root of the characteristic equation of Refs. [3, 16]; the model reproduces both, and the two-quadrature roots of Ref. [16], to machine precision. The same model therefore predicts the recorded response in the wideband, where the loop dominates, and in the beam band, where the coupled mode does.

F. Parameter classes

The physics fixes the structure and the parameters of Table I. The beam current of each record is measured; the loop gain is $G = G_0\,10^{(\texttt{COUNTS.A}-0.9\,\mathrm{dB})/20}$, with G_0 the gain at the setting of 2 October, because gain tracking holds the forward path constant. The remaining parameters are identified, at the level at which physics says they can vary:

- station constants, common to all records: G_0, τ, Q_L, the measurement-chain coefficients and, where stated, α_c;
- per operating point (date and current): the compression g_A and the noise angle α; the AM/PM conversion k_{AP}, which the closed-loop records do not determine (Sec. IV), is fixed at the value measured in the trip records, and the alternative of one compression for all currents is tested in Sec. V;
- per group of records taken minutes apart at one current: the detuning Δf;
- per record: the loop phase θ, which carries the unrecorded tracking correction.

The steady generator-current angle θ_g follows from the detuning and the current.

IV. IDENTIFIABILITY

A. What one record can and cannot determine

In the wideband the beam changes the recorded response by a few percent; the response is set by the loop. For a complex-linear loop the five parameters G, θ, τ, Q_L and Δf are determined by one record: at 22 mA a single record returns them with statistical errors of 0.004 in G, 0.1° in θ, 2 ns in τ, 35 in Q_L and 0.15 kHz in Δf. A compressed klystron and a one-dimensional excitation change this. For cavities on tune and no beam, $\mathbf{M} = m(s)\mathbf{A}$ with the scalar $m = GcZ/R_L$ and the constant matrix $\mathbf{A} = \mathbf{R}(\theta)\mathbf{K}$. With $\mathbf{A} = \mathbf{V}\mathrm{diag}(\lambda_1, \lambda_2)\mathbf{V}^{-1}$, Eq. (8) becomes

$$H(f) = -\sum_{i=1,2} w_i(\alpha)\,\frac{m(s)\lambda_i}{1 + m(s)\lambda_i}, \tag{9}$$

a weighted sum of two closed-loop responses whose loop gains are the eigenvalues of $\mathbf{A}$. The four quantities θ, g_A, k_{AP}, α (with G) are carried by two complex gains and their weights, and the shapes of the two terms differ only through their bandwidths $(1 + \lambda_i)\omega_{1/2}$. A single complex-linear response with an effective gain and phase approximates such a sum closely, which is why a linear model with effective parameters fits each record (Sec. V). Detuning and beam break the symmetry only partially. Moreover, the phase-path gain of the klystron and the controller gain enter only as a product, and the synchrotron frequency enters the wideband only through the beam's small contribution there.

B. What operating-condition diversity adds

A change of `PHASE.C` rotates $\mathbf{A}$ by a known angle, and a change of `COUNTS.A` scales it by a known factor; neither changes g_A, k_{AP} or α at a given operating point. Each additional record at the same operating point therefore adds about 1400 data points but only one parameter, θ, and moves the two eigenvalues of Eq. (9) in a prescribed way. Records at other beam currents share the station constants and constrain the detuning through the beam loading.

Figure 2 quantifies the effect for the compression of the klystron at 497 mA. The profile likelihood of g_A is computed with every other parameter re-optimized [22]. From one record in the routine's own band, g_A lies between 0.29 and 0.76 (95%), a linear klystron is disfavoured by only $\Delta\chi^2 = 7$, k_{AP} is unconstrained, and the normalized Fisher information has condition number 2.1×10^6. Adding the beam band of the same record narrows g_A to 0.30–0.38 ($\Delta\chi^2 = 98$ at $g_A = 1$). The seven records of the 497 mA gain and phase scan give 0.33–0.35 ($\Delta\chi^2 = 466$), and all eighteen records 0.35–0.36 ($\Delta\chi^2 = 1967$). The compression is therefore identified, and robustly so: its value moves by less than 0.04 between data sets. The AM/PM coefficient is not. It is bounded within each data set ($k_{\mathrm{AP}} = -0.08$, 95% interval -0.17 to $+0.02$, from the scan; -0.44, -0.47 to -0.42, from all records), but it moves between data sets together with the loop phase at 497 mA, because a rotation of the loop and an amplitude-to-phase conversion in the klystron produce similar changes of the projected response. The closed-loop records therefore identify the compression but not the AM/PM conversion, which has to come from elsewhere.

C. What the trip records add

The trip records measure the klystron without any loop model (App. C). In the normal operation before each trip, the relative fluctuations of the klystron output amplitude follow those of the drive amplitude with a ratio of 0.40 ± 0.05 between 50 and 200 kHz over the 10 records (Fig. 3a): 0.42 ± 0.02 at full output and 0.39 ± 0.06 at the no-beam output. The coherence is 0.93–0.98, and the H_2 estimate, biased the other way by noise, gives 0.41, so the ratio is the klystron's small-signal compression g_A. Below 10 kHz it is 0.44, and above 200 kHz it falls to 0.34 ± 0.04, where the two measurement chains need not be flat. The probe's response to drive amplitude, relative to its response to drive phase, gives the AM/PM conversion: for a memoryless klystron and any plant that treats both quadratures alike the ratio is $k_{\mathrm{AP}} - jg_A$, and the records

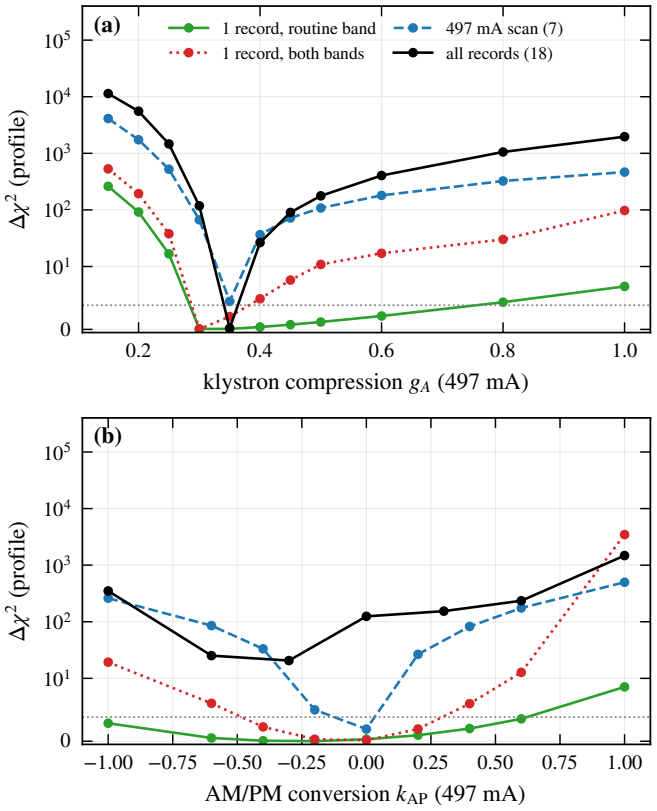


FIG. 2. Profile likelihood of the klystron's (a) amplitude compression g_A and (b) AM/PM conversion k_{AP} at the 497 mA operating point, with both free at every operating point, for nested data sets: one record in the routine's band (15–400 kHz), the same record with the beam band, the gain and phase scan at 497 mA, and all records. Every other parameter is re-optimized at each point; the dotted line marks the 95% level ($\Delta\chi^2 = 3.84$). The vertical axis is linear below 10.

give $k_{AP} = -0.10 \pm 0.05$ (Fig. 3b). The loop's own correlation pulls this ratio towards a linear klystron—the same regression returns $g_A = 0.50$ instead of 0.40—so $|k_{AP}|$ is if anything underestimated, by about a fifth. The values that the closed-loop records give with k_{AP} free at every operating point, from -0.44 to $+0.62$, lie far outside this range.

We therefore fix the AM/PM conversion at the trip-record value. This raises χ^2 by 351, 1.4 % of its value, moves the compression at full current by at most 0.05, and leaves the nuisance parameters regular: the noise angle, which depends only on the klystron's insertion phase, varies by 17° between operating points instead of 71°.

V. COMPETING EXPLANATIONS

A. The anomaly

Fitted record by record with a complex-linear loop, every record is described at the noise level, but the parameters drift with the operating point (Fig. 5). Normalized to the same gain setting, the effective loop gain falls from 0.9 at 22 mA to 0.4 at 500 mA; the effective delay rises from 625 to 700 ns; the effective Q_L falls from about 6800 to 4500–5300. None of these is plausible for the hardware: the gain is held by gain tracking, the delay is cable and klystron transit time, and Q_L is fixed by the couplers and agrees at low current with the value from the reflected power. At 497 mA the effective gain also scales more steeply than the gain setting, and at 22 mA it changes with the phase setting—0.91 to 0.86 for `PHASE.C` from −14° to −23°—which no rotation can do. The same records show the coupled beam mode as a resonance near $\pm f_s$ whose frequency falls from 10.28 to about 8.0 kHz with current, and whose damping rate rises from about 1.3×10^3 to $2.4 \times 10^3\,\mathrm{s}^{-1}$ as the gain is lowered by 1.5 dB at 497 mA (Fig. 4).

B. Explanations and their test

Table III compares the candidate explanations, each fitted to all eighteen records in both bands (25200 real data points).

- *One station, linear klystron.* Shared hardware, detuning from the data, beam dynamics included. It fails by more than an order of magnitude in both bands: the change of the response with current is not a beam-dynamics effect at fixed hardware.

- *Unequal cavity detunings.* A spread of the four detunings broadens the vector-sum response and could mimic a lower Q_L. Fitted freely it reaches spreads of tens of kilohertz, which the measured reflected power of every cavity (1–3%) excludes, and it still fails.

- *Effective parameters per record.* Per-record G, τ and Q_L describe the wideband at the noise level but not the beam band: the damping of the coupled mode they imply is too small. They also lack a physical reading and do not transfer between conditions (Sec. VI).

- *Compressed klystron.* Shared hardware, the klystron's compression and the noise angle per operating point, its AM/PM conversion from the trip records. With the momentum compaction identified (Sec. VI A) it describes both bands at the noise level with 45 parameters; with the design value it fails in the beam band and pulls the station constants away from their measured values.

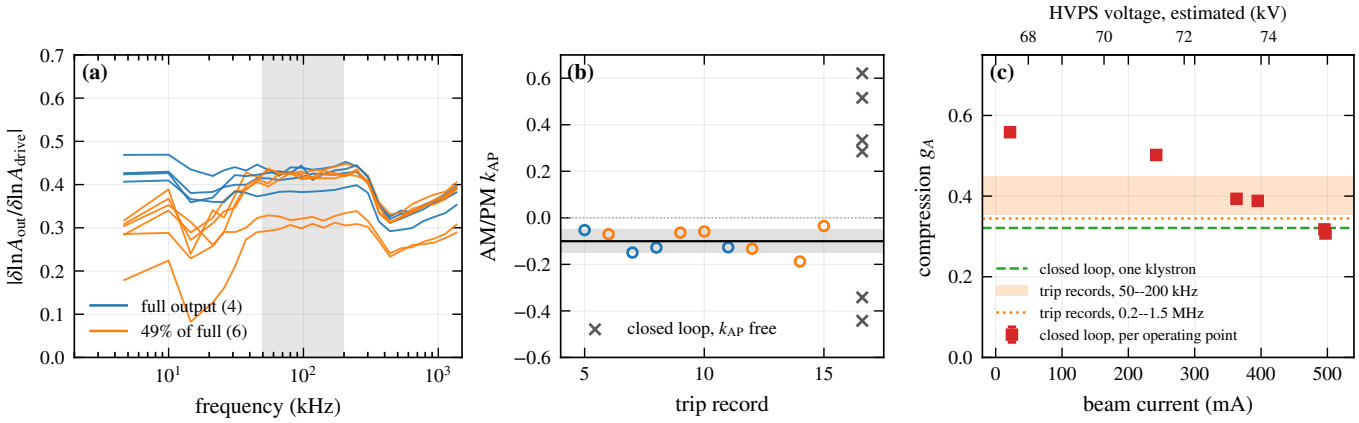


FIG. 3. The klystron in the trip records (present tube, 2021). (a) Ratio of the relative fluctuations of output and drive amplitude against frequency, for the records at full output (blue) and at the no-beam output (orange); shaded: 50 kHz to 200 kHz. (b) AM/PM conversion from the probe's response to drive amplitude and phase, per record (band: mean ± standard deviation); crosses: the values of the closed-loop fit with k_{AP} free at every operating point. (c) Compression identified from the closed-loop records at each operating point (squares) and as one value for all currents (dashed), against the trip records at 50 kHz to 200 kHz (band) and above 200 kHz (dotted); top axis: HVPS voltage estimated from the power balance at the fixed drive.

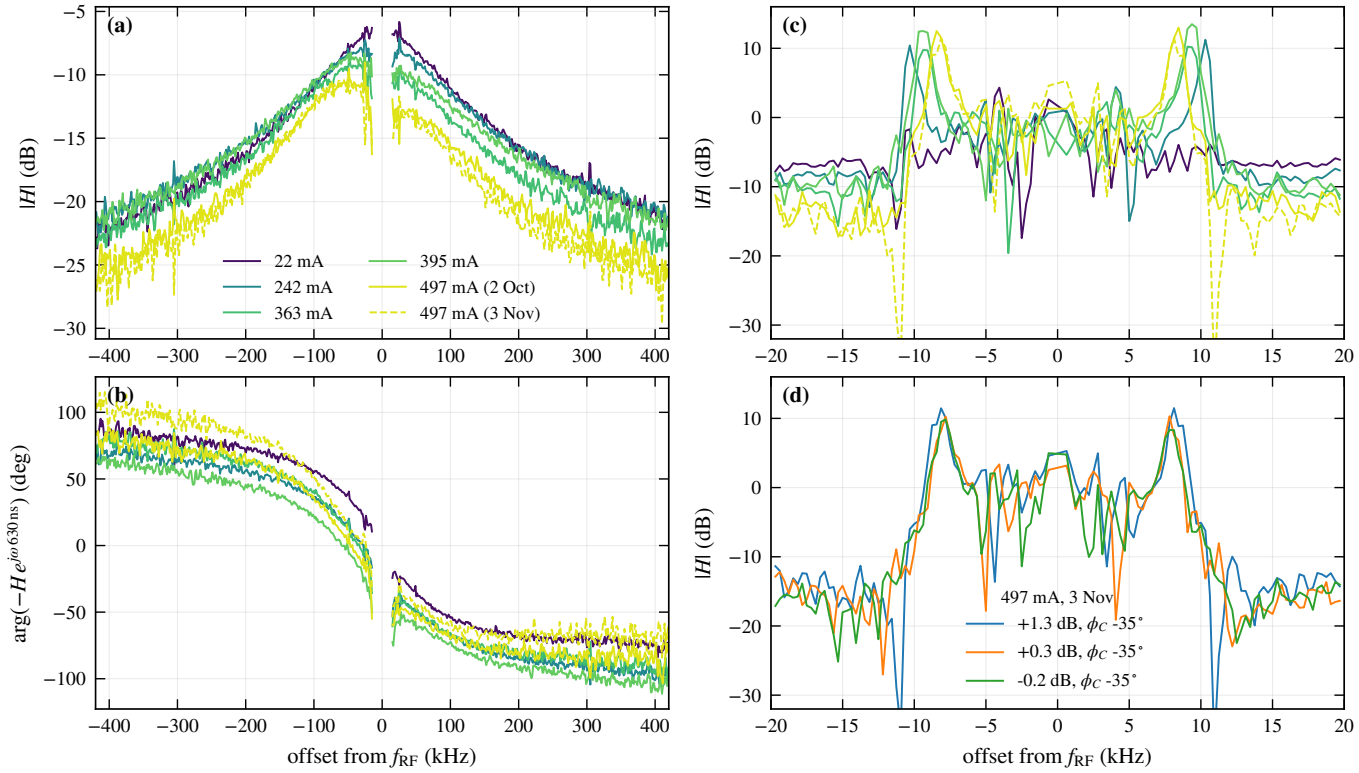


FIG. 4. Measured closed-loop responses. (a) Magnitude and (b) phase in the wideband, the phase with 630 ns of delay removed; colour: beam current. (c) The beam band at 312 Hz resolution for the same records: the coupled mode appears as a resonance near $\pm f_s$ that moves down with current. (d) The beam band at 497 mA for three gain settings.

Variants test the klystron itself. Its compression and AM/PM taken from the trip records, with no klystron parameter fitted, already describe the records far better than any linear model ($\chi^2/\nu = 1.176$). One compression for all currents is worse (ΔAIC = 2132), and freeing the AM/PM conversion at every operating point lowers χ^2 only by the amount discussed in Sec. IV C, with values that the trip records exclude. Without the beam's dipole motion the fit fails by far (+13912).

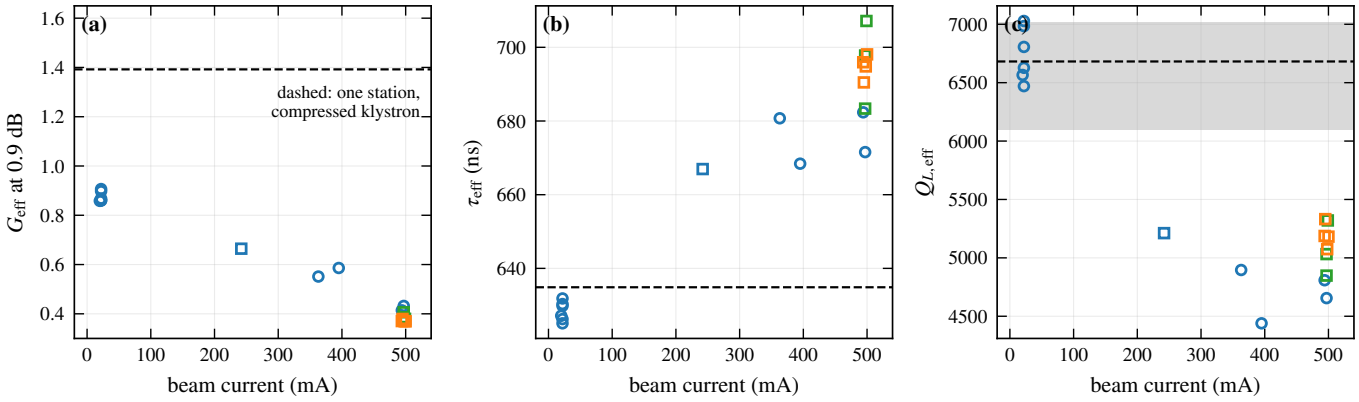


FIG. 5. Effective parameters of record-by-record complex-linear fits: (a) loop gain normalized to the 2 October gain setting, (b) loop delay, (c) loaded Q. Circles: 2 October and 3 November (242 mA); squares: 3 November at 497 mA, the gain scan in orange (+1.3 dB in green). Dashed: the station constants of the selected model (Sec. V C); grey band in (c): Q_L of the four cavities from their reflected power.

Figure 6(b) shows where the models fail. The linear station misses everywhere; the effective model, the single klystron and the selected model are close to the noise level above 100 kHz; below 100 kHz the selected model leaves residuals up to twice the noise, concentrated in the two extreme phase settings at 22 mA (χ^2/ν of 2.1–2.5 in their wideband) and in two of the 497 mA records. These are the records in which the per-record effective gain and Q_L differ most from the rest, and the structure that remains is a few percent of $|H|$.

C. The selected model

The station constants are $G_0 = 1.39 \pm 0.01$, $\tau = 635 \pm 1$ ns and $Q_L = 6681 \pm 12$ (statistical errors). The delay agrees with the 625 ns measured on the station, and Q_L with the value from the cavities' reflected power (6760, cavity range 6100–7020). The measurement chain contributes an odd magnitude slope of 0.77 dB and an even phase of 3.7° at ±400 kHz, common to all records. Table IV lists the operating-point parameters. The detunings follow the beam-loading compensation, $\Delta f \approx -36\,\text{kHz} \times I_b/500\,\text{mA}$, to within about 3 kHz, as the tuners' load-angle offset allows. The loop phase at 22 mA follows `PHASE.C` with a slope of 1.24 degrees per degree, as found independently in Ref. [16].

The klystron's compression grows with the beam current, from $g_A = 0.56$ at 22 mA through 0.50, 0.39, 0.39 at 242, 363 and 395 mA to 0.31–0.32 at 494–497 mA, and the two high-current operating points, measured a month apart, agree within their errors. This is the trend that the station's operation implies: the drive is the same at every current, but with beam the tube runs at a higher voltage and much closer to saturation (efficiency 47 % against 32 % without beam, Sec. II). The HVPS voltage was not recorded with the closed-loop records. Estimated from the power balance at the fixed drive—cavity losses, beam power and the reflection at $\beta \approx 3.8$, scaled by 0.96 to the snapshots and interpolated between their voltages—it rises from about 67 kV at 22 mA through 71, 73, 74 kV to 75 kV at 494–497 mA (Table IV), and the compression falls monotonically with it. The phase path of the loop therefore has the gain $G_0 = 1.39$ at the 2 October setting, while a perturbation of the drive amplitude meets a loop gain of only $g_A G_0 \approx 0.43$ at 500 mA. The "loop gain of 0.88" obtained earlier from the same routine at low current [16] is neither of the two but their projection on the direction of the injected noise [Eq. (8)].

VI. VALIDATION

A. Out-of-band prediction of the coupled mode

The first test asks whether the loop identified in the wideband alone predicts the coupled mode, which lives in a band that was not used. For each of the 12 records with more than 200 mA, the coupled mode is measured without a model by fitting a pole pair and a smooth background to both sides of the beam band (App. B); frequencies are determined to about 29 Hz and damping rates to about 173 s^{-1}. The models are then evaluated with no adjustment.

Both wideband models fail, in opposite ways (Fig. 8a). The effective linear model places the mode at about the right frequency (mean error +198 Hz) but with too little damping (mean error −836 s^{-1}, rms 897 s^{-1}): its effective parameters describe the projection that the routine measures, not the impedance that the beam sees. The compressed-klystron model gets the damping roughly right (mean error −289 s^{-1})—the klystron's compression supplies the damping that a linear model lacks, as conjectured in Ref. [16]—but places the mode −841 Hz too low. The prediction thus isolates one discrepancy, the coherent frequency, in a model whose loop is otherwise

TABLE III. Candidate explanations fitted to all records in the wideband and the beam band. The compressed-klystron models use the AM/PM conversion of the trip records unless stated. n_p: free parameters; χ^2/ν overall and per band; ΔAIC and ΔBIC relative to the selected model [23, 24].

Model	n_p	χ^2/ν	wideband	beam band	ΔAIC	ΔBIC
One station, linear klystron	33	15.462	16.79	5.56	363542	363445
+ unequal cavity detunings	41	14.937	16.13	5.97	350222	350189
Effective G, τ, Q_L per record	83	1.406	0.95	4.71	9838	10147
Klystron of the trip records, none fitted	39	1.176	1.21	0.92	4004	3955
Compressed klystron, one for all currents	40	1.101	1.14	0.79	2132	2091
Per operating point, design α_c	44	1.250	1.16	1.86	5891	5883
Per operating point, no beam motion	45	1.569	1.11	4.89	13912	13912
Per operating point, AM/PM free	51	1.002	1.04	0.70	−339	−291
Selected: compression per operating point	45	1.016	1.06	0.71	0	0

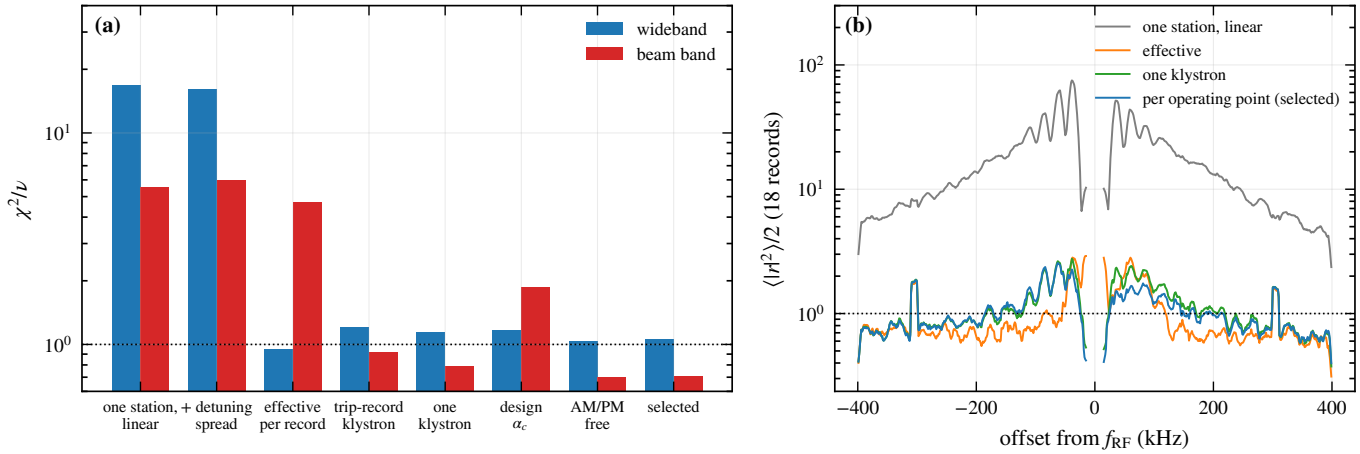


FIG. 6. (a) χ^2/ν of the candidate explanations of Table III in the wideband and the beam band (all records); the last three have the compression per operating point. (b) Normalized squared residuals in the wideband, averaged over all records and over nine neighbouring bins.

TABLE IV. Operating-point parameters of the selected model: klystron compression g_A (statistical errors), noise angle α in the cavity frame, and detuning Δf of each group of records. The AM/PM conversion is fixed at $k_{AP} = -0.10$ (trip records, Sec. IV C). V_k: HVPS voltage, not recorded, estimated from the power balance at the fixed drive (Sec. V C).

Op. point	n	V_k (kV)	g_A	α (°)	Δf (kHz)
2 Oct 22 mA	6	67	0.56 ± 0.01	−11	−2.0, +0.8
2 Oct 363 mA	1	73	0.39 ± 0.01	−15	−24.6
2 Oct 395 mA	1	74	0.39 ± 0.01	−21	−28.4
2 Oct 494 mA	2	75	0.32 ± 0.01	−11	−32.7
3 Nov 242 mA	1	71	0.50 ± 0.01	−10	−19.3
3 Nov 497 mA	7	75	0.31 ± 0.01	−5	−32.9

right.

The coherent frequency at fixed loop depends on the beam through ω_s^2 and $\kappa_1 I_{b1}$, both proportional to α_c. The mode at 242 mA, at 10.28 kHz, already lies above the nominal incoherent frequency $f_s = 10.03$ kHz, whereas in every model considered here the reactive loading of cavities tuned below f_{RF} pulls the coherent mode below f_s. Identifying α_c jointly with the loop, from both bands, gives $\alpha_c = 1.412 \times 10^{-3}$, with a statistical error of 0.003×10^{-3} and a range of 1.37–1.44×10^{-3} over the compressed-klystron models of Table III that leave it free, i.e., an incoherent synchrotron frequency of 10.97 kHz at 2.85 MV (10.8–11.1 kHz), 9% above the nominal value. With the design value the fit worsens in both bands ($\chi^2/\nu = 1.16$ and 1.86 against 1.06 and 0.71), and the station constants move away from their measured values ($Q_L = 6184$, $\tau = 645$ ns). Two alternatives were tested. Freeing the beam-coupling strength separately from ω_s returns a coupling factor 0.97 ± 0.01 with $\alpha_c = 1.40 \times 10^{-3}$: both scale as α_c does, which a calibration error of the gap voltage (acting on ω_s alone) would not produce. And the effective linear model does not improve with α_c free: its deficit is the damping. With α_c identified the coupled mode agrees with the model-free poles to 80 Hz in frequency and 336 s^{-1} in damping (rms over 12 records), across currents from 242 to 500 mA and gain settings from −0.2 to +1.3 dB (Fig. 8); with one compression for all currents the damping comes out +482 s^{-1} too high on average. The identified value lies 20–28% above the published values, 1.10–1.18×10^{-3} [13, 16]; it

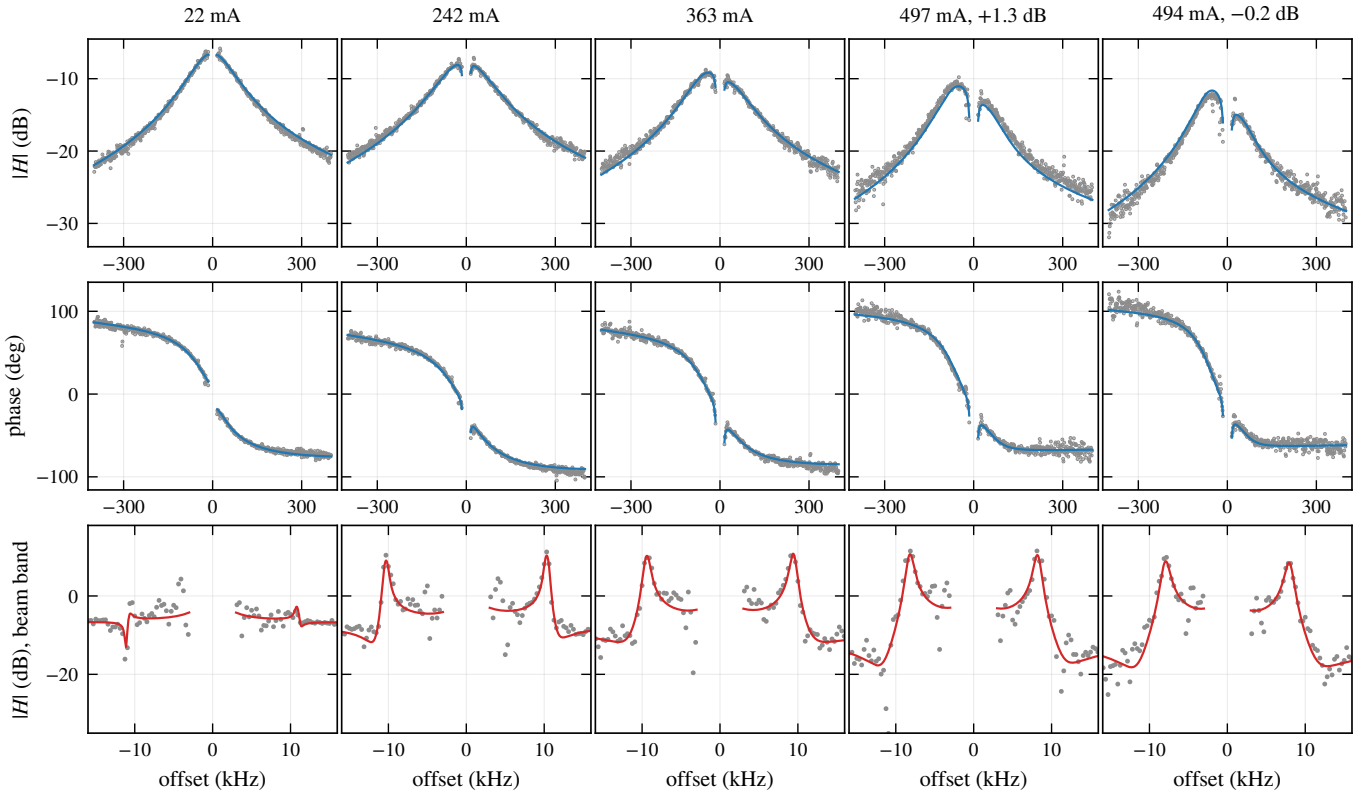


FIG. 7. Recorded responses (grey) and the selected model (lines) for five operating points: magnitude and phase in the wideband (top, middle; phase with 630 ns removed) and magnitude in the beam band (bottom, model convolved with the window kernel). One set of station constants describes all records.

is either a property of the present lattice or a proxy for an unmodelled element that acts on the coupled mode like a stronger synchrotron focusing. A direct measurement of the incoherent synchrotron frequency at a few milliamperes, where the coherent shift vanishes, decides between the two: the model predicts 10.97 kHz.

B. Prediction across operating conditions

The second test asks whether the identified model predicts records taken under conditions it was not trained on. Each model is refitted without the held-out records; for these only nuisance parameters are refitted: the loop phase, plus the detuning (and the noise angle, for the klystron models) when the beam current is new. The effective model needs its per-record parameters at the new condition: they are taken from the same operating point and scaled with the gain setting (gain transfer), averaged over the neighbouring phase settings (phase transfer), or interpolated in beam current (current transfer); the selected model interpolates g_A in beam current, and the single klystron needs nothing. Figure 9 shows the held-out χ^2/ν.

For a new beam current the selected model predicts the record almost as well as it fits it in sample: 0.92, 0.78 and 0.69 at 242, 363 and 395 mA, against 0.79, 0.74 and 0.63 in sample; the single klystron gives 1.32, 0.94 and 0.71, and interpolated effective parameters give 2.65, 6.34 and 10.31. For new gain settings at 497 mA the selected model gives 1.37 on average (in sample 0.83) and the scaled effective model 1.91, the difference lying in the beam band (0.6–0.7 against 5–6.5). For new phase settings at 22 mA both give about 2 (2.09 and 2.22), close to the selected model's in-sample value of these two records (2.06). Freeing the AM/PM conversion at every operating point improves no prediction (0.95, 0.82, 0.63; 1.35; 2.04). The one-station linear model fails every test.

C. Comparison with the trip records

At full current the identified compression, 0.31–0.32 at 494–497 mA (0.35 with the AM/PM conversion free), lies below the trip-record value at full output, 0.42 ± 0.02 at 50 kHz to 200 kHz, and close to the trip-record value above 200 kHz, 0.34 ± 0.04 (Fig. 3c). Both describe a strongly compressed klystron: a change of the drive amplitude reaches the output with a third to two fifths of its relative size. At low output the two disagree. The trip records give 0.39 ± 0.06 at the no-beam output, where the closed-loop records give 0.56 at 22 mA; in the trip records

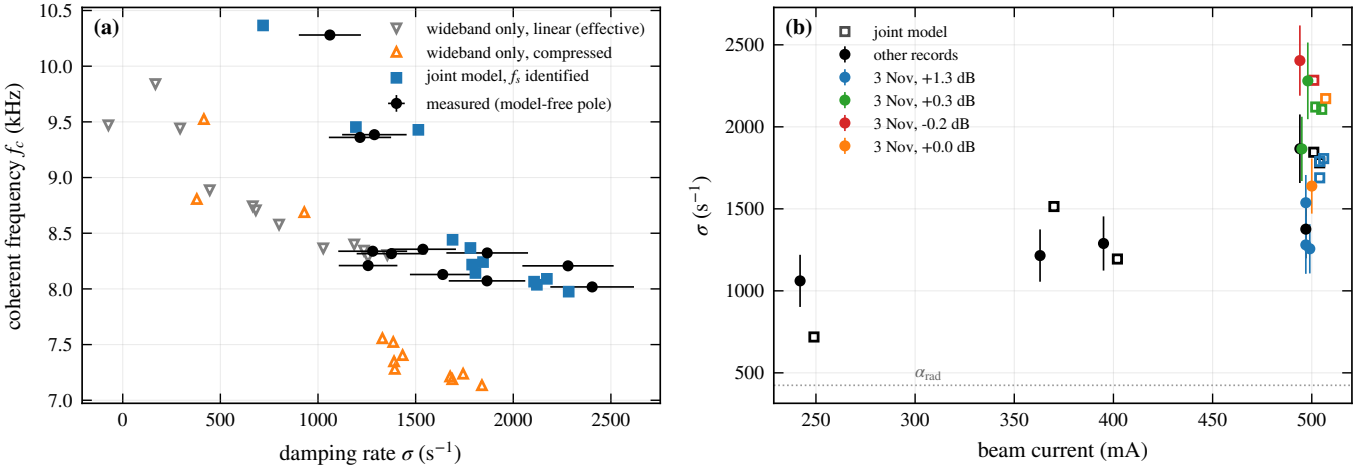


FIG. 8. The coupled beam–cavity–klystron–loop mode. (a) Frequency and damping rate measured without a model (black, 1σ errors), predicted from the wideband alone by the effective linear model (grey) and by the compressed-klystron model with nominal ring parameters (orange), and given by the joint model with the synchrotron frequency identified (blue). (b) Damping rate against beam current; at 497 mA the colours mark the gain setting. Dotted: radiation damping.

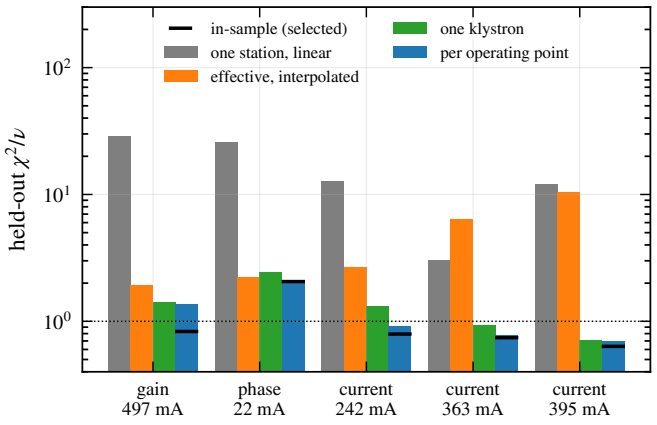


FIG. 9. Held-out χ^2/ν (both bands) of records predicted from models trained without them: gain transfer at 497 mA (four records), phase transfer at 22 mA (two), and three beam currents. Bars: one-station linear model, effective model with transferred parameters, one compressed klystron for all currents, compression per operating point (selected). Black ticks: in-sample values of the selected model.

the compression hardly depends on the output, in the closed-loop records it does. A single compression for all currents, 0.32, costs $\Delta\chi^2 = 2142$: 1124 from the 22 mA records, 834 from the single records at 242–395 mA, and 184 from the eleven records at 494–497 mA. The current dependence thus rests on the low-current phase scan, which no model reproduces to better than twice the noise in its extreme settings, and on the records at intermediate current, among them the beam band at 242 and 363 mA, where a single compression overestimates the damping. It is the trend expected from the klystron's efficiency at the two voltages (Sec. V C). The trip records are five years older, and their beam currents, cathode voltages and loop settings were not recorded; a transfer-curve measurement at the present drive and at the cathode voltages of low and high current would settle the point. Taken as they are, with no klystron parameter fitted, the trip-record values describe all eighteen records better than any linear model ($\chi^2/\nu = 1.176$, Table III).

D. Physical consistency

The station constants agree with independent knowledge: τ with the delay measured on the station, Q_L with the reflected power, the detunings with the beam-loading compensation, the gain constant across operating points with the gain tracking, the loop-phase slope at low current with the earlier calibration, and the growth of the compression with the klystron's efficiency. None of these was imposed.

VII. DISCUSSION

a. What changes with the beam. The dominant change of the loop dynamics between 20 and 500 mA is the compression of the klystron, which runs closer to saturation as the HVPS raises its voltage to supply the beam's power at a fixed drive; the detuning and the beam's dipole motion are the other two current-dependent elements, and both follow from physics once the loop is known. A complex-linear model can follow the change only with effective parameters that mean nothing physical, and such a model does not predict the coupled mode or new operating points. The compression divides the loop into two: a phase loop of gain $G_0 \approx 1.39$ and an amplitude loop of about one third of that at 500 mA ($g_A G_0 \approx 0.43$). The weaker amplitude loop leaves more

of the cavity's natural impedance to the beam, and that supplies the Robinson damping that a linear model with the installed integral network cannot explain [16]. Lowering the gain by 1.5 dB at 497 mA raises the measured damping rate from about 1.4×10^3 to $2.4 \times 10^3\,\mathrm{s}^{-1}$; the model reproduces the direction and about half the size of this change (1.8×10^3 to $2.3 \times 10^3\,\mathrm{s}^{-1}$), but at the highest gain it overestimates the damping by about $404\,\mathrm{s}^{-1}$, 2.4 times the measurement error.

b. The synchrotron frequency. The beam band identifies a combination of beam parameters that the loop band cannot, and the result disagrees with the design value by 9% in f_s. Because the incoherent synchrotron frequency is easy to measure at low current, this is a falsifiable prediction; if it is confirmed, the momentum compaction of the present lattice should be revised in every longitudinal model of the ring; if not, the coupled-mode band contains dynamics that the model lacks, most likely in the integral path or the klystron at a few kilohertz.

c. Limits. The model assumes rigid bunches and a uniform fill, equal vector-sum weights and a memoryless klystron, and it treats the measurement chain as a smooth common factor. The loop phase of each record is a free parameter because the tracking correction was not recorded. The AM/PM conversion is taken from trip records five years older than the closed-loop data, because the closed-loop records determine it only together with the high-current loop phase; its value is small, and the compression at full current hardly depends on it. The low-current phase scan is reproduced only to about twice the noise, operating-point parameters for currents with a single record rest on that record, and the HVPS voltages of the operating points between 0 and 500 mA are estimated, not measured.

d. Measurements that would add most. A two-dimensional excitation—independent noise sequences on I and Q, or a multisine whose lines sit on one side of the carrier—would measure H_1 and H_2 separately and determine the AM/PM conversion from the closed-loop records themselves. The klystron's transfer curves (AM/AM and AM/PM) at the present drive and at the cathode voltages of low and high current would test the identified growth of the compression with current, which the older trip records do not show. Recording the HVPS voltage and the klystron's drive and output power with every closed-loop record would tie each record to its point on those curves, and recording the total loop phase (`PHASE.L`) would remove one nuisance parameter per record. The incoherent synchrotron frequency at low current settles the question of α_c. With these, the same routine records would give a complete, predictive model of the station—and, for the replacement LLRF system, whose loop delay is longer, a quantitative basis for setting its gain and phase.

VIII. CONCLUSIONS

Operational closed-loop measurements, taken with an existing diagnostic routine and a one-dimensional excitation, identify a compact physics-informed model of the coupled beam–cavity–klystron–feedback system of the SPEAR3 RF station. One set of station constants—loop gain, delay, loaded Q—describes eighteen records from 20 to 500 mA, at several loop gains and phases, on two days a month apart, once the klystron is allowed to compress its amplitude response as it approaches saturation; the constants agree with independent station data. Single records do not determine the klystron's asymmetry; records at several loop settings and the beam-mode band determine its compression, and trip records of the same tube supply its small AM/PM conversion, which the closed-loop records leave correlated with the loop phase. The compression grows with the beam current as the tube, held at a fixed drive, works closer to saturation; at full current it agrees with the trip records to within their spread across frequency bands, while the trip records show less dependence on the output power. Explanations by effective linear parameters, by unequal cavity detunings, or by beam dynamics at fixed hardware are rejected by the data. The identified model predicts records at beam currents it was not trained on almost at in-sample quality, where interpolated effective parameters fail. Its out-of-band prediction of the coupled beam mode isolated the one beam parameter that the loop band cannot determine; identified from the beam band, it implies an incoherent synchrotron frequency 9% above the design value, whatever the klystron model, a prediction that a simple low-current measurement can test. The physics supplied the structure; the data determined the klystron's compression, the station constants and the synchrotron focusing, and tested every explanation against measurements it was not built from.

ACKNOWLEDGMENTS

We thank the SSRL operations and RF groups for the closed-loop records, the trip records and the station snapshots on which this study rests. All measurements analyzed here were taken by existing operational routines; no setting of the station was changed for this work.

DATA AND CODE AVAILABILITY

The closed-loop records, trip records and station snapshots, the analysis code (model, estimators, fits, validation) and the scripts that generate every figure and every number quoted in the text are kept in the study folder *Coupled-System Identification* of the SPEAR3 RF repository; the model reproduces the characteristic-equation and two-quadrature results of Ref. [16] to machine precision (test script included).

Appendix A: Beam-loaded plant and recorded response

With the cavity voltages $\boldsymbol{v}_k = \mathbf{Z}_k(\boldsymbol{i}_g - I_{b1}\hat{\boldsymbol{b}}\,\Delta\phi)$ and Eq. (2),

$$\left(s^2 + 2\alpha_{\rm rad}s + \omega_s^2\right)\Delta\phi = \kappa_1\hat{\boldsymbol{p}}^{\mathsf{T}}\mathbf{Z}_\Sigma\boldsymbol{i}_g - \kappa_1 I_{b1}\left(\hat{\boldsymbol{p}}^{\mathsf{T}}\mathbf{Z}_\Sigma\hat{\boldsymbol{b}}\right)\Delta\phi, \tag{A1}$$

so $\Delta\phi = \kappa_1\hat{\boldsymbol{p}}^{\mathsf{T}}\mathbf{Z}_\Sigma\boldsymbol{i}_g/D(s)$ and $\boldsymbol{v}_m = \mathbf{Z}_m\boldsymbol{i}_g - I_{b1}\mathbf{Z}_m\hat{\boldsymbol{b}}\,\Delta\phi$ gives Eq. (3). The term $\kappa_1 I_{b1}\hat{\boldsymbol{p}}^{\mathsf{T}}\mathbf{Z}_\Sigma\hat{\boldsymbol{b}}$ equals $-j\kappa[Z_\Sigma(s) - \bar{Z}_\Sigma(s)]$ with $\kappa = \kappa_1 I_{b1}/2$, so that $D(s)$ is the open-loop characteristic function of Refs. [2, 16]; multiplied by the cavity denominator it is Pedersen's quartic.

The loop of Eq. (6) places the klystron between the drive chain and the cavities. Blocks with a real impulse response commute with every 2×2 matrix, so only the order of the rotation $\mathbf{R}(\theta)$, the plant $\mathbf{P}_b$ and the klystron $\mathbf{K}$ matters. The noise is injected into the drive in the RFP frame; with ψ the phase from the RFP drive frame to the cavity frame, conjugating every signal by $\mathbf{R}(\psi)$ moves the loop to the cavity frame and turns the injection angle $\alpha_{\rm RFP}$ into $\alpha = \alpha_{\rm RFP} + \psi$. The loop phase θ includes ψ and the controller rotation; a change of `PHASE.C` changes θ but not α, while a change of the klystron's insertion phase with its cathode voltage changes both.

If the excitation is one-dimensional, with a real amplitude $r(t)$ along a fixed angle α, $\boldsymbol{n}(t) = r(t)(\cos\alpha, \sin\alpha)^{\mathsf{T}}$, the complex input is $n = e^{j\alpha}r$ and $\overline{n(-f)} = e^{-2j\alpha}n(f)$; the widely linear output $x = H_1 n + H_2\overline{n(-f)}$ is then $(H_1 + H_2 e^{-2j\alpha})\,n$, Eq. (8). The recorded response is fully coherent with the reference, but it is one projection: H_1 and H_2 cannot be separated from one record, and parameters that act only through the image response H_2—the klystron's asymmetry above all—are confounded with the loop parameters (Sec. IV).

The closed-loop poles are the zeros of $\det[\mathbf{1} + \mathbf{M}(s)]D(s)$. With $\mathbf{F} = \mathbf{K}\,Gc\,\mathbf{R}(\theta)/R_L$, Sylvester's identity gives $\det(\mathbf{1}+\mathbf{M}) = \det(\mathbf{1}+\mathbf{F}\mathbf{P}_b)$, and since $\mathbf{P}_b$ is $\mathbf{Z}_m$ minus a rank-one term, the matrix determinant lemma gives the exact factorization

$$\det[\mathbf{1} + \mathbf{M}(s)]\,D(s) = \det(\mathbf{1} + \mathbf{F}\mathbf{Z}_m)\,D_0(s), \tag{A2}$$

where $D_0(s) = s^2 + 2\alpha_{\rm rad}s + \omega_s^2 + \kappa_1 I_{b1}\hat{\boldsymbol{p}}^{\mathsf{T}}\mathbf{Z}_b\hat{\boldsymbol{b}}$ is the characteristic function of the beam with the closed-loop impedance $\mathbf{Z}_b = \mathbf{Z}_\Sigma(\mathbf{1} + \mathbf{F}\mathbf{Z}_m)^{-1}$ used in Ref. [16], and $\det(\mathbf{1} + \mathbf{F}\mathbf{Z}_m)$ that of the RF loop without beam. The coupled mode is tracked by Newton continuation from $I_b = 0$, where it is the incoherent oscillator $s = -\alpha_{\rm rad} \pm j(\omega_s^2 - \alpha_{\rm rad}^2)^{1/2}$.

Appendix B: Estimator statistics

The H_1 estimate with output noise only is unbiased; its complex error variance is $\sigma_H^2 = |H|^2(1-\gamma^2)/(\gamma^2 n_{\rm eff})$ [17], with $n_{\rm eff} = n/(1 + 2c^2)$ and $c = 1/6$ the correlation of Hann windows overlapping by half. Two independent estimates from the two halves of each record test this calibration: for a calibrated complex Gaussian error the ratio $|\hat{H}_a - \hat{H}_b|^2/(2\sigma^2_{H,\rm half})$ has median $\ln 2 = 0.693$; the records give 0.75 (range 0.65–0.88). The mean of the same ratio exceeds one because a few percent of the bins carry interference lines (harmonics near ±25, ±50 and −305 kHz) whose scatter the smoothed variance underestimates. Values of χ^2/ν slightly below one therefore indicate fits at the noise level of the bulk of the bins, and differences of χ^2 between models are compared on the same bins.

In the beam band the Welch estimate is, for a flat excitation spectrum, the transfer function convolved with the normalized power kernel $|W(f)|^2$ of the Hann window; the models are evaluated on a four-times oversampled grid over ±4 bins and convolved with that kernel. The model-free estimate of the coupled mode fits, on both sides of the carrier at once, a smooth background (complex quadratic in f) plus the residues of one complex pole pair $s = -\sigma \pm j\omega_c$, convolved in the same way.

Appendix C: Trip records

At a trip the RFP and IQ&A memories freeze at the fault, and the IOC writes them to files. Each of the 15 records of 2021, all of the present tube, holds 51 200 samples per signal at 10.24 MS/s, the 5.0 ms before the fault, stored newest first. The RFP signal memory holds the klystron drive (multiplexer code 3 in the two low bits of each word) and the cavity memory one cavity probe (code 3, cavity 4). Each IQ&A module records a single channel at the full rate; for module 1 it is channel 0, the klystron output, whose amplitude $(I^2 + Q^2)^{1/2}$ is linear in the RF amplitude. As for the closed-loop records, the stored I/Q are the complex conjugate of the physical envelope: conjugated, the drive-to-probe response places the cavities below $f_{\rm RF}$ in every record, its phase-path response being 1.2–6.3 times larger at −45 kHz to −20 kHz than at 20 kHz to 45 kHz. In 10 records every memory holds RF; in the others the RFP memories, or all of them, are empty. The analysis uses the normal operation up to 0.5 ms before the end of each record, earlier if any signal leaves its level first.

The compression is the cross-spectrum of the relative output amplitude with the relative drive amplitude over the drive's auto-spectrum (H_1; 2048-point Hann segments with 50 % overlap), with the ratio of the output's auto-spectrum to the cross-spectrum (H_2) as the bound from the other side; the two modules differ in timing by 189 ns, which only turns the phase of the ratio. For the AM/PM conversion the relative fluctuation c of the probe is regressed, bin by bin (1024-point segments), on the relative drive amplitude a_d and the drive phase φ_d: $c = A\,a_d + B\,\varphi_d$. For a memoryless klystron $c = P[(g_A + jk_{\rm AP})\,a_d + j\varphi_d]$ for any complex-linear plant P, so that $A/B = k_{\rm AP} - jg_A$ whatever the

cavities and their detuning; the beam acts on the probe only near $\pm f_s$, outside the band of 60 kHz to 250 kHz used. Feedback correlates the drive with disturbances of the probe and pulls the ratio towards that of a linear klystron ($A/B \to -j$); the compression the regression returns, 0.50 against 0.40 measured directly, gauges that pull.

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